

TryHackMe - Investigating with ELK 101

ElasticStack Overview

Elastic stack

Elastic stack is the collection of different open source components linked together to help users take the data from any source and in any format and perform a search, analyze and visualize the data in real-time.



Elastic Search



Logstash



Beats



Kibana

Elasticsearch

Elasticsearch is a full-text search and analytics engine used to store JSON-formatted documents. Elasticsearch is an important component used to store, analyze, perform correlation on the data, etc. Elasticsearch supports RESTful API to interact with the data.

Logstash

Logstash is a data processing engine used to take the data from different sources, apply the filter on it or normalize it, and then send it to the destination which could be Kibana or a listening port. A logstash configuration file is divided into three parts:

- The **input** part is where the user defines the source from which the data is being ingested. Logstash supports many input plugins as shown in the reference <https://www.elastic.co/guide/en/logstash/8.1/input-plugins.html>
- The **filter** part is where the user specifies the filter options to normalize the log ingested above. Logstash supports many filter plugins as shown in the reference

documentation <https://www.elastic.co/guide/en/logstash/8.1/filter-plugins.html>

- The **Output** part is where the user wants the filtered data to send. It can be a listening port, Kibana Interface, elasticsearch database, a file, etc. Logstash supports many Output plugins as shown in the reference

documentation <https://www.elastic.co/guide/en/logstash/8.1/output-plugins.html>

Beats

Beats is a host-based agent known as Data-shippers that is used to ship/transfer data from the endpoints to elasticsearch. Each beat is a single-purpose agent that sends specific data to the elasticsearch. All available beats are shown below.

The Beats family

All kinds of shippers for all kinds of data.

Filebeat

Lightweight shipper for logs and other data



Metricbeat

Lightweight shipper for metric data



Packetbeat

Lightweight shipper for network data



Winlogbeat

Lightweight shipper for Windows event logs



Auditbeat

Lightweight shipper for audit data



Heartbeat

Lightweight shipper for uptime monitoring



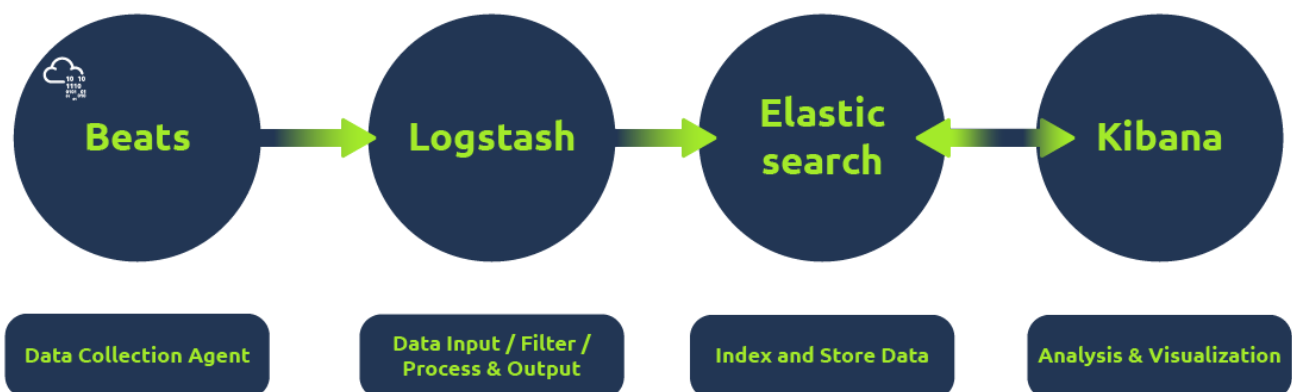
Functionbeat

Serverless shipper for cloud data



Kibana

Kibana is a web-based data visualization that works with elasticsearch to analyze, investigate and visualize the data stream in real-time. It allows the users to create multiple visualizations and dashboards for better visibility.



Discover Tab

Kibana Discover tab is a place where analyst spends most of their time. This tab shows the ingested logs (also known as documents), the search bar, normalized fields, etc. Here analysts can perform the following tasks:

- Search for the logs
- Investigate anomalies
- Apply filter based on
- Search term
- Time period

Discover Tab

Discover tab within the Kibana interface contains the logs being ingested manually or in real-time, the time-chart, normalized fields, etc. Analysts use this tab mostly to search/investigate the logs using the search bar and filter options.



Some key information available in a dashboard interface are:

1. **Logs (document):** Each log here is also known as a single document containing information about the event. It shows the fields and values found in that document.
2. **Fields pane:** Left panel of the interface shows the list of the fields parsed from the logs. We can click on any field to add the field to the filter or remove it from the search.
3. **Index Pattern:** Let the user select the index pattern from the available list.

4. **Search bar:** A place where the user adds search queries / applies filters to narrow down the results.
5. **Time Filter:** We can narrow down results based on the time duration. This tab has many options to select from to filter/limit the logs.
6. **Time Interval:** This chart shows the event counts over time.
7. **TOP Bar:** This bar contains various options to save the search, open the saved searches, share or save the search, etc.